
Attempt all questions

1. Let $(G, *)$ be a group. Prove that for any $x, y \in G$,

$$(xy)^{-1} = y^{-1}x^{-1}.$$

2. Let $(G, *)$ be a group. Show that the equations

$$ax = b \quad \text{and} \quad ya = b$$

have solutions $x, y \in G$ for all choices of $a, b \in G$.

3. Let $G = \{a, b\}$ be equipped with the binary operation $*$: $G \times G \rightarrow G$ defined by

$$(a, a) \mapsto a, \quad (a, b) \mapsto a, \quad (b, a) \mapsto b, \quad (b, b) \mapsto b.$$

Is $(G, *)$ a group? Justify your answer.

4. Let $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$. Prove that $(\mathbb{R}^+, \times)$ is a group. Prove that $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \times)$ are isomorphic.

5. Let G be a group and $y \in G$. Prove that the map

$$\phi : G \rightarrow G, \quad x \mapsto y^{-1} * x * y$$

is an isomorphism.

6. Let G be a group and $\text{Aut}(G)$ be the set of all automorphisms of G . Observe that the composition of two automorphisms is again an automorphism. Prove that the composition of functions makes $\text{Aut}(G)$ a group.

Best wishes